```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
import seaborn as sns
```

```
data = pd.read_csv("/content/Boston House Price Data.csv")
data
```

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	PRICE
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1.0	296.0	15.3	396.90	4.98	24.0
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2.0	242.0	17.8	396.90	9.14	21.6
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2.0	242.0	17.8	392.83	4.03	34.7
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3.0	222.0	18.7	394.63	2.94	33.4
4	0.06905	0.0	2.18	0.0	0.458	7.147	54.2	6.0622	3.0	222.0	18.7	396.90	5.33	36.2
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
501	0.06263	0.0	11.93	0.0	0.573	6.593	69.1	2.4786	1.0	273.0	21.0	391.99	9.67	22.4
502	0.04527	0.0	11.93	0.0	0.573	6.120	76.7	2.2875	1.0	273.0	21.0	396.90	9.08	20.6
503	0.06076	0.0	11.93	0.0	0.573	6.976	91.0	2.1675	1.0	273.0	21.0	396.90	5.64	23.9
504	0.10959	0.0	11.93	0.0	0.573	6.794	89.3	2.3889	1.0	273.0	21.0	393.45	6.48	22.0
505	0.04741	0.0	11.93	0.0	0.573	6.030	80.8	2.5050	1.0	273.0	21.0	396.90	7.88	11.9

506 rows × 14 columns

```
data.head()
```

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	PRICE
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1.0	296.0	15.3	396.90	4.98	24.0
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2.0	242.0	17.8	396.90	9.14	21.6
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2.0	242.0	17.8	392.83	4.03	34.7
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3.0	222.0	18.7	394.63	2.94	33.4
4	0.06905	0.0	2.18	0.0	0.458	7.147	54.2	6.0622	3.0	222.0	18.7	396.90	5.33	36.2

```
data.shape
```

```
(506, 14)
```

```
data.isnull()
```

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	PRICE
0	False	False	False	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
501	False	False	False	False	False	False	False	False	False	False	False	False	False	False
502	False	False	False	False	False	False	False	False	False	False	False	False	False	False
503	False	False	False	False	False	False	False	False	False	False	False	False	False	False
504	False	False	False	False	False	False	False	False	False	False	False	False	False	False
505	False	False	False	False	False	False	False	False	False	False	False	False	False	False

506 rows × 14 columns

```
data.isnull().sum()
```

	0
CRIM	0
ZN	0
INDUS	0
CHAS	0
NOX	0
RM	0
AGE	0
DIS	0
RAD	0
TAX	0
PTRATIO	0
B	0
LSTAT	0
PRICE	0

dtype: int64

```
data.duplicated()
```

	0
0	False
1	False
2	False
3	False
4	False
...	...
501	False
502	False
503	False
504	False
505	False

506 rows × 1 columns

dtype: bool

```
data.columns
```

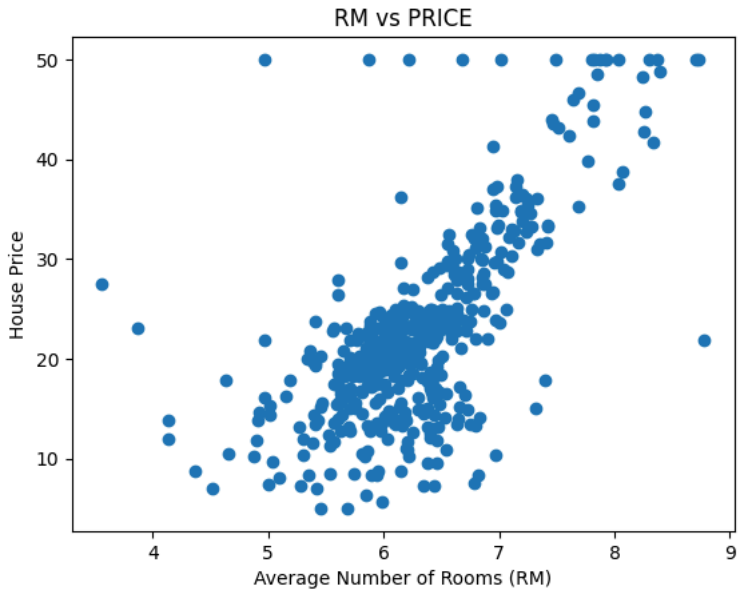
```
Index(['CRIM', 'ZN', 'INDUS', 'CHAS', 'NOX', 'RM', 'AGE', 'DIS', 'RAD', 'TAX',
       'PTRATIO', 'B', 'LSTAT', 'PRICE'],
      dtype='object')
```

```
data.empty
```

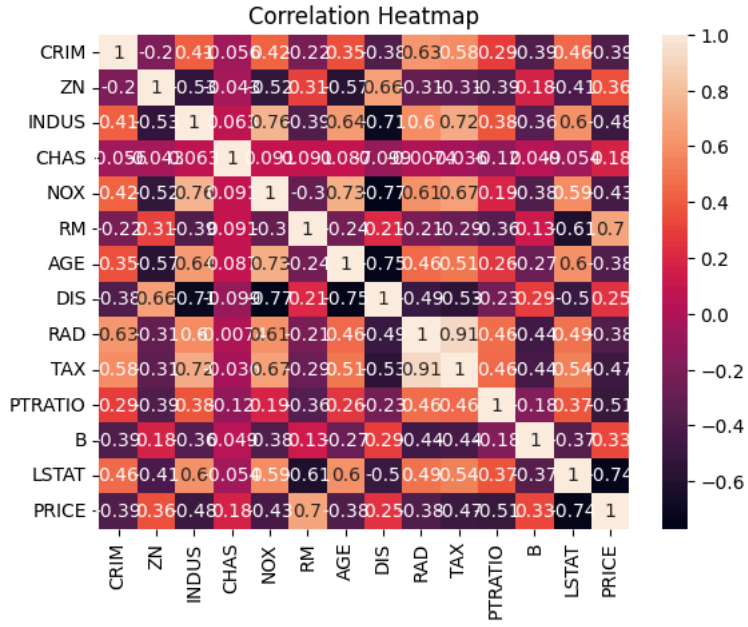
```
False
```

```
import matplotlib.pyplot as plt

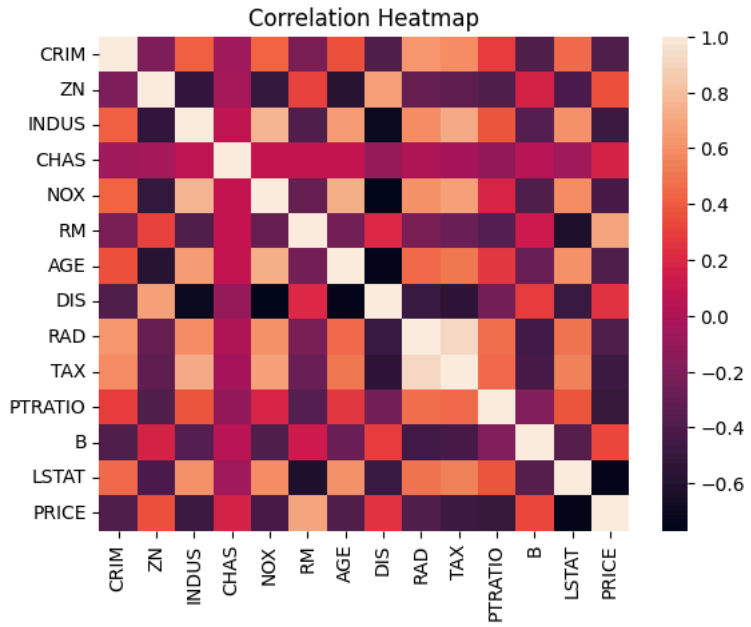
plt.scatter(data["RM"],data["PRICE"])
plt.xlabel("Average Number of Rooms (RM)")
plt.ylabel("House Price")
plt.title("RM vs PRICE")
plt.show()
```




```
correlation = data.corr(numeric_only=True)
sns.heatmap(correlation, annot=True)
plt.title("Correlation Heatmap")
plt.show()
```




```
correlation = data.corr(numeric_only=True)
sns.heatmap(correlation, annot=False)
plt.title("Correlation Heatmap")
plt.show()
```




```
from sklearn.model_selection import train_test_split

#import features
X= data.drop("PRICE",axis=1)

#Target Variable
Y= data["PRICE"]

#Split dataset into training and testing data
X_train, X_test,Y_train, Y_test = train_test_split(X,Y,test_size=0.2, random_state=42)

print("Trainind data:",X_train.shape)
print("Testing data:",X_test.shape)
```

```
Trainind data: (404, 13)
Testing data: (102, 13)
```

```
print("Training Data:")
print(X_train.head())
```

```
Training Data:
         CRIM    ZN  INDUS  CHAS     NOX  ...   RAD    TAX  PTRATIO       B  LSTAT
477  15.02340   0.0  18.10   0.0  0.6140  ...  24.0  666.0     20.2  349.48  24.91
15    0.62739   0.0   8.14   0.0  0.5380  ...   4.0  307.0     21.0  395.62   8.47
332   0.03466  35.0   6.06   0.0  0.4379  ...   1.0  304.0     16.9  362.25   7.83
423   7.05042   0.0  18.10   0.0  0.6140  ...  24.0  666.0     20.2    2.52  23.29
19    0.72580   0.0   8.14   0.0  0.5380  ...   4.0  307.0     21.0  390.95  11.28

[5 rows x 13 columns]
```

```
print("Testing Data:")
print(X_test.head())
```

```
Testing Data:
        CRIM    ZN  INDUS  CHAS    NOX  ...   RAD    TAX  PTRATIO       B  LSTAT
173  0.09178   0.0   4.05   0.0  0.510  ...   5.0  296.0     16.6  395.50   9.04
274  0.05644  40.0   6.41   1.0  0.447  ...   4.0  254.0     17.6  396.90   3.53
491  0.10574   0.0  27.74   0.0  0.609  ...   4.0  711.0     20.1  390.11  18.07
72   0.09164   0.0  10.81   0.0  0.413  ...   4.0  305.0     19.2  390.91   5.52
452  5.09017   0.0  18.10   0.0  0.713  ...  24.0  666.0     20.2  385.09  17.27

[5 rows x 13 columns]
```

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train, Y_train)
print("Model Trained Successfully")
```

```
Model Trained Successfully
```

```
model.fit(X_train, Y_train)
```

```
▾ LinearRegression
LinearRegression()
```

```
model.predict(X_test)
```

```
array([28.99672362, 36.02556534, 14.81694405, 25.03197915, 18.76987992,
       23.25442929, 17.66253818, 14.34119   , 23.01320703, 20.63245597,
       24.90850512, 18.63883645, -6.08842184, 21.75834668, 19.23922576,
       26.19319733, 20.64773313,  5.79472718, 40.50033966, 17.61289074,
       27.24909479, 30.06625441, 11.34179277, 24.16077616, 17.86058499,
       15.83609765, 22.78148106, 14.57704449, 22.43626052, 19.19631835,
       22.43383455, 25.21979081, 25.93909562, 17.70162434, 16.76911711,
       16.95125411, 31.23340153, 20.13246729, 23.76579011, 24.6322925 ,
       13.94204955, 32.25576301, 42.67251161, 17.32745046, 27.27618614,
       16.99310991, 14.07009109, 25.90341861, 20.29485982, 29.95339638,
       21.28860173, 34.34451856, 16.04739105, 26.22562412, 39.53939798,
       22.57950697, 18.84531367, 32.72531661, 25.0673037 , 12.88628956,
       22.68221908, 30.48287757, 31.52626806, 15.90148607, 20.22094826,
       16.71089812, 20.52384893, 25.96356264, 30.61607978, 11.59783023,
       20.51232627, 27.48111878, 11.01962332, 15.68096344, 23.79316251,
        6.19929359, 21.6039073 , 41.41377225, 18.76548695,  8.87931901,
       20.83076916, 13.25620627, 20.73963699,  9.36482222, 23.22444271,
       31.9155003 , 19.10228271, 25.51579303, 29.04256769, 20.14358566,
       25.5859787 ,  5.70159447, 20.09474756, 14.95069156, 12.50395648,
       20.72635294, 24.73957161, -0.164237  , 13.68486682, 16.18359697,
       22.27621999, 24.47902364])
```

```
Y_Pred = model.predict(X_test)
```

```
print("Prediction Values:",Y_Pred[0:3])
```

```
Prediction Values: [28.99672362 36.02556534 14.81694405]
```

```
from sklearn.metrics import  mean_squared_error,r2_score

mse = mean_squared_error(Y_test,Y_Pred)
```